

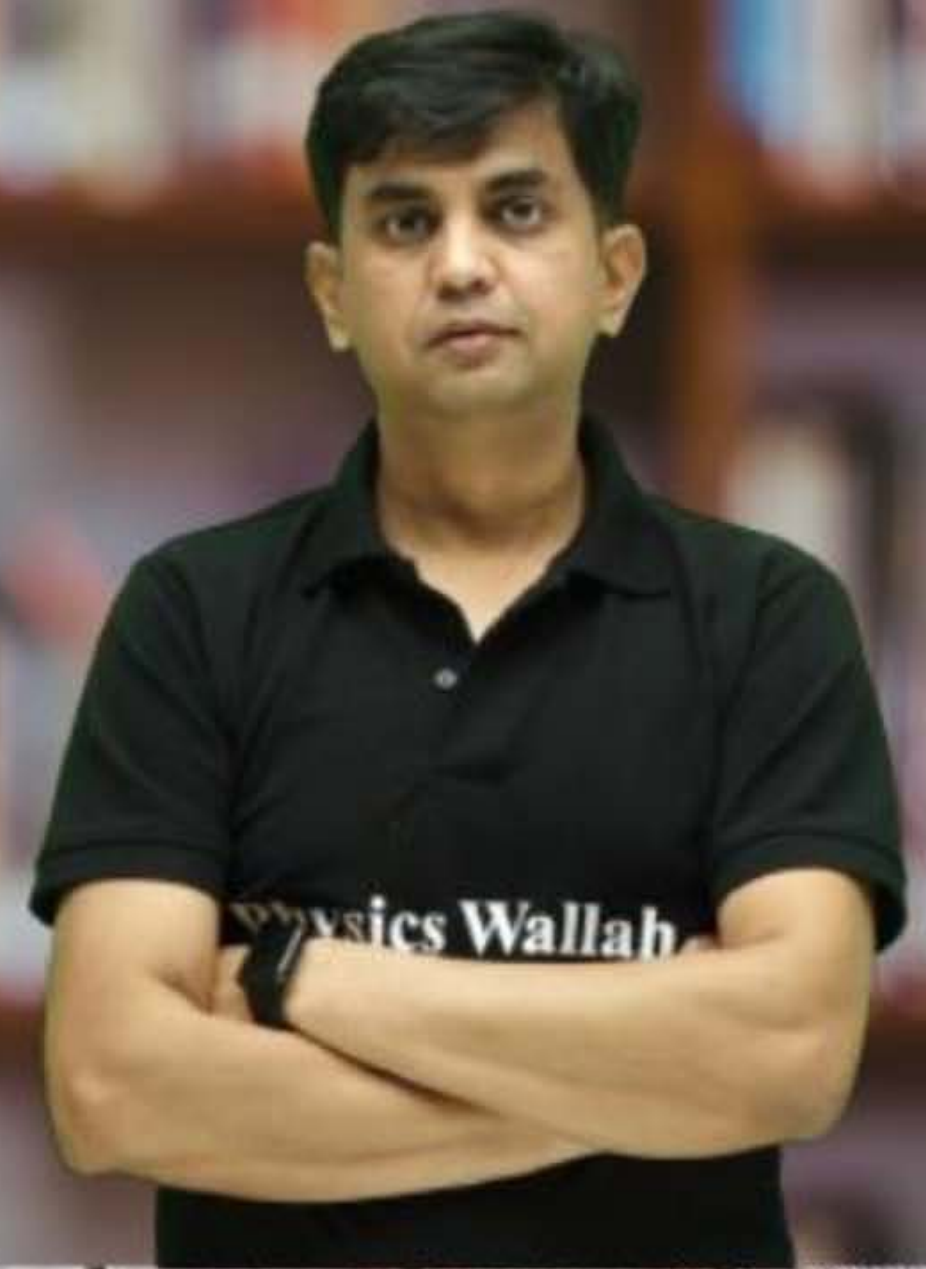
CS & IT ENGINEERING



Computer Network

Application Layer

Lecture No. - 02



By - Abhishek Sir



Recap of Previous Lecture



Topic

DNS



Topics to be Covered



Topic

DNS

Topic

HTTP

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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[NAT]

[GATE-2022][1 Mark]



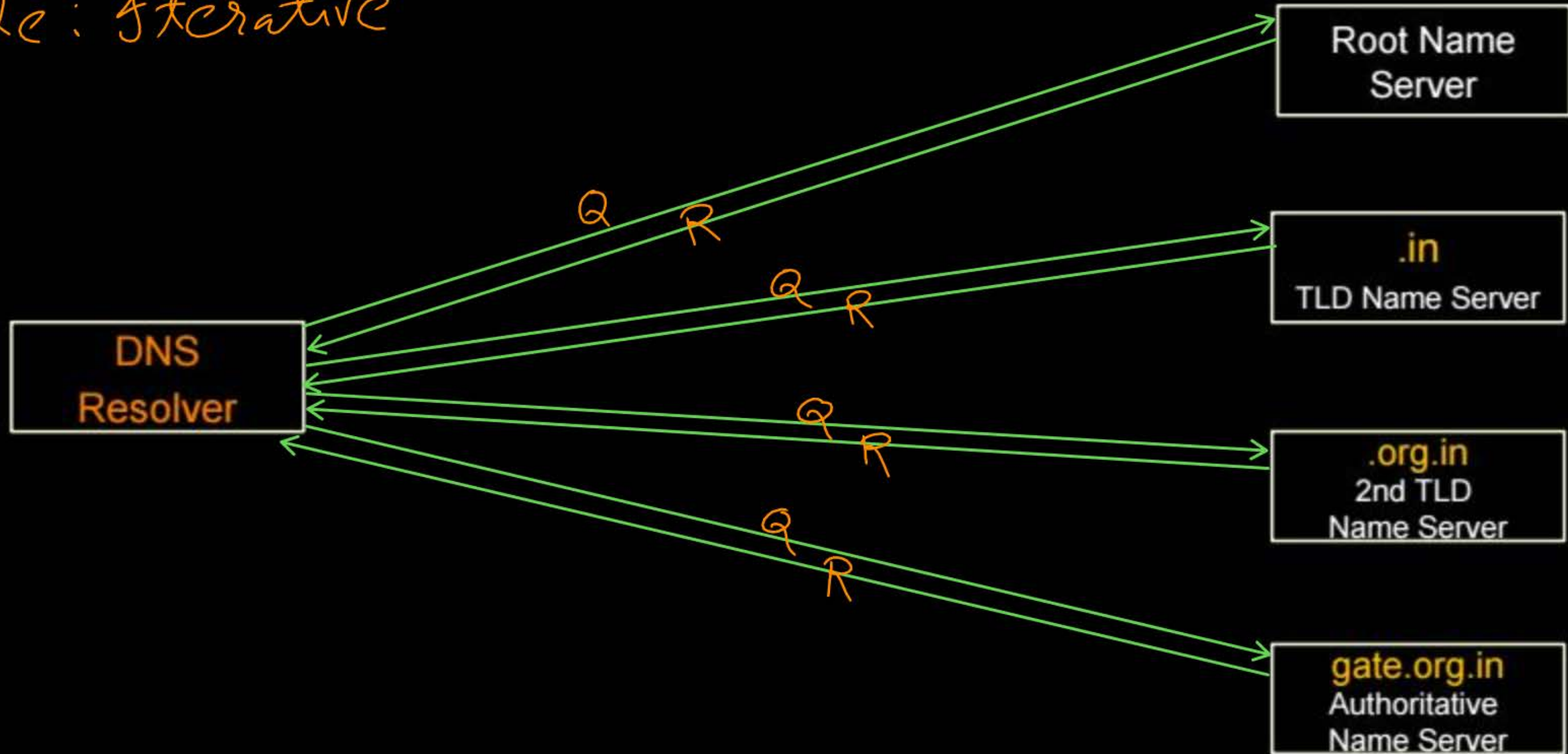
#Q. Consider the resolution of the domain name `www.gate.org.in` by a DNS resolver. Assume that no resource records are cached anywhere across the DNS servers and that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is _____.

Ans: 4

Domain Name : "www.gate.org.in"

Mode : Iterative

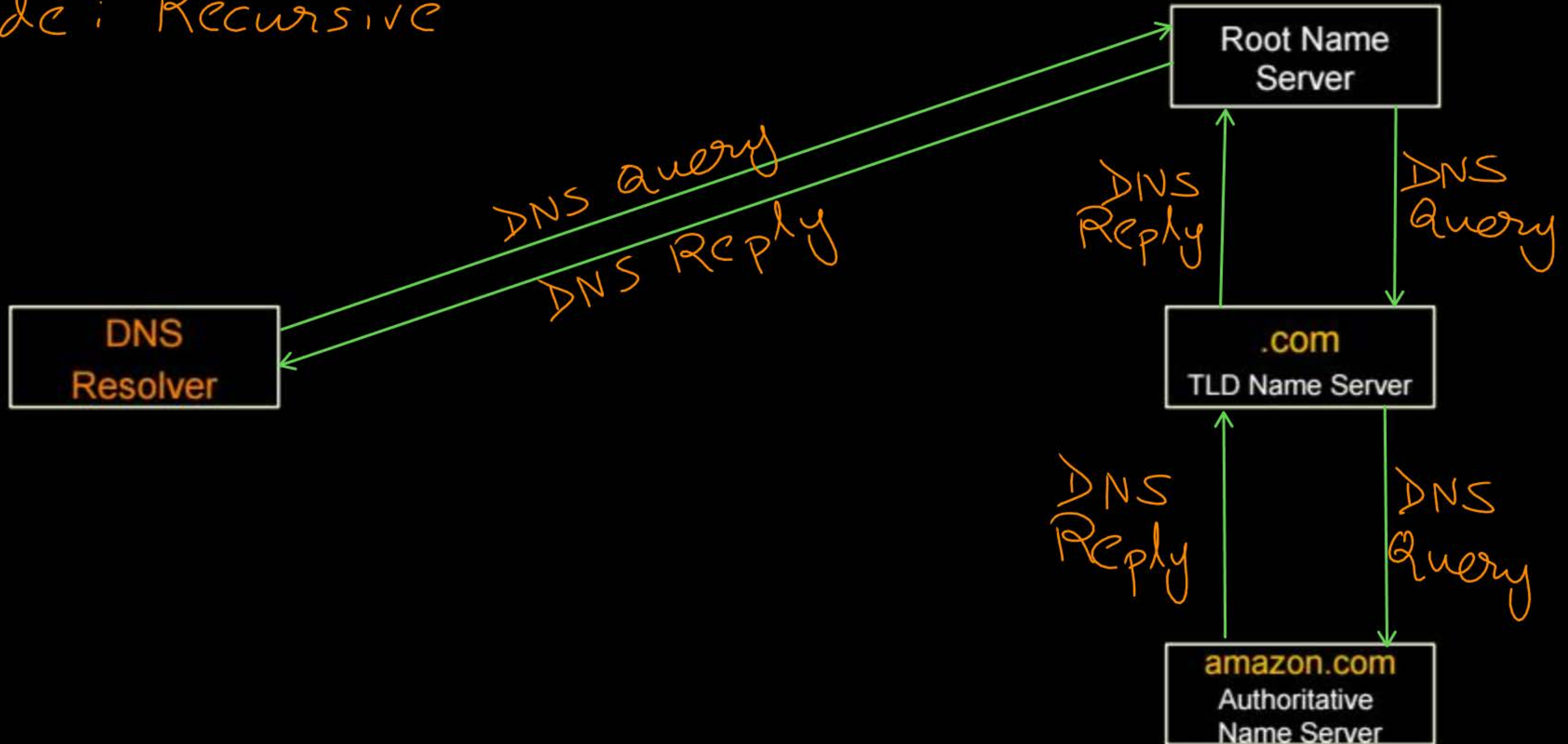
DNS Hierarchy



Domain Name : "www.amazon.com"

Mode : Recursive

DNS Hierarchy





Topic : DNS Caching



- Once (any) name server learns mapping, it caches mapping
- Immediately returns a cached mapping in response to a query
- => Caching improves response time
- => Cached entries time-out (disappear) after some time (TTL)



Topic : DNS Records



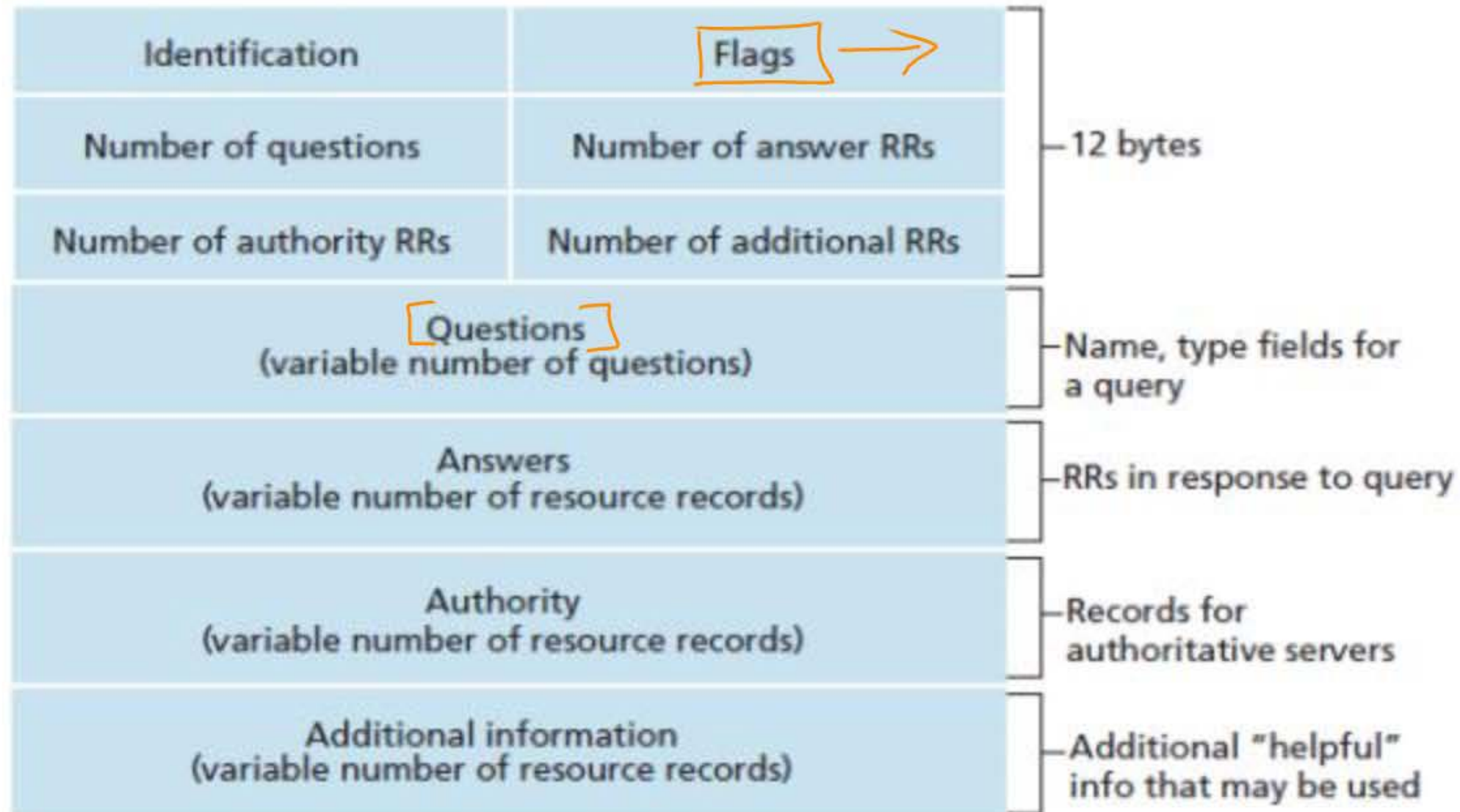
- **Distributed database** storing **resource records** (**RR**)
- **RR format** : (Name, Value, Type, TTL)

Type	Name	Value
A	Host Name	IP Address
NS	Domain Name	Authoritative Name Server IP Address
CNAME	Alias Name	Canonical Name
MX	Name	Associated SMTP Mail Server Name



Topic : DNS Message Format

→ DNS query and reply message have same format





Topic : DNS



→ DNS uses UDP as transport protocol for query and reply message when message size is less than equal to 512 bytes

(Default)

→ if DNS messages size is greater than 512 bytes then DNS uses TCP as transport protocol

→ if DNS client doesn't get response from DNS server for its query then DNS uses TCP as transport protocol for query and reply message

→ DNS uses TCP for 'zone transfer' ✓



Topic : Web Page



→ Web page consists of objects

[Object can be HTML file, JPEG image, Java Applet, Audio File etc]
↓
Text file

→ Each objects can be stored on different web servers

→ Web page consists of base HTML file which includes several referenced objects

[Each addressable by a URL]



Topic : HTTP



- Hypertext Transfer Protocol : (HTTP) ✓
- Web's application layer protocol
- Client - Server Model



Topic : HTTP is stateless



- Server maintains no information about past client requests
[Every request from client will be treated as new request by server]
- Protocols that maintain "state" are complex ✓
[Past history (state) must be maintained]



Topic : HTTP



Client :

→ Browser that requests, receives, (using HTTP protocol)
and display web objects

Server :

→ Web server sends (using HTTP protocol) objects in response to requests



Topic : HTTP uses TCP



→ Server creates TCP socket on Port 80 (default port number)

→ Client initiates TCP connection to server

→ Server accepts TCP connection from client

} TCP connection
Establishment

→ HTTP messages (application layer protocol messages) exchanged
between browser (HTTP client) and web server (HTTP server)

→ TCP connection closed



Topic : HTTP Connections



→ Two types of HTTP connections:

1. Non-persistent HTTP : HTTP/1.0
2. Persistent HTTP : HTTP/1.1



2 mins Summary



Topic

DNS ✓

Topic

HTTP



THANK - YOU

